EXPERIMENT NO. 10

Title: Design and Implementation of a Real-Time Traffic Light Control System Using an Embedded System

1. Aim

To design and implement a real-time traffic light control system using an Arduino-based embedded system to control red, yellow, and green LEDs in a predefined sequence with appropriate time delays.

2. Theory

An embedded system is a combination of hardware and software designed to perform a specific task. A traffic light control system uses a microcontroller to control traffic signals automatically according to predefined timing intervals.

Working Principle:

The Arduino Uno acts as the controller and operates three LEDs representing traffic signals.

- Red Light: Indicates STOP.
- Yellow Light: Indicates WAIT or prepare to stop.
- Green Light: Indicates GO.

The microcontroller switches the LEDs ON and OFF in a predefined sequence using programmed time delays. This provides automatic and continuous operation without manual intervention.

Sequence of Operation:

Red → Green → Yellow → Repeat

3. Hardware and Software Requirements

Hardware:

1. Arduino Uno
2. Red LED
3. Yellow LED
4. Green LED

5. 220 Ω resistors
6. Breadboard
7. Jumper wires
8. USB cable

Software:

1. Arduino IDE
2. Embedded C/C++ programming
3. Serial Monitor (optional)

4. Algorithm / Procedure

1. Connect the red, yellow, and green LEDs to digital pins 8, 9, and 10 of the Arduino Uno, respectively, using suitable current-limiting resistors.
2. Connect the LED cathodes to GND.
3. Open the Arduino IDE and write the program for traffic light control.
4. Configure the LED pins as output pins using `pinMode()`.
5. Turn ON the red LED for 5 seconds and keep the other LEDs OFF.
6. Turn ON the green LED for 5 seconds and keep the other LEDs OFF.
7. Turn ON the yellow LED for 2 seconds and keep the other LEDs OFF.
8. Repeat the sequence continuously using the `loop()` function.
9. Upload the program to the Arduino Uno and observe the LED sequence.

5. Program Logic

None

```
START
  |
Initialize LED pins
  |
Turn ON RED LED
Wait 5 seconds
  |
Turn ON GREEN LED
Wait 5 seconds
  |
Turn ON YELLOW LED
Wait 2 seconds
  |
Repeat the sequence
```

algorithm

6. Observation

The following observations are expected during the execution of the program:

Sr. No.	Red LED	Yellow LED	Green LED	Duration
1	ON	OFF	OFF	5 seconds
2	OFF	OFF	ON	5 seconds
3	OFF	ON	OFF	2 seconds
4	ON	OFF	OFF	Sequence repeats

Observations:

1. The red LED glows for 5 seconds, indicating STOP.
2. The green LED glows for 5 seconds, indicating GO.
3. The yellow LED glows for 2 seconds, indicating WAIT.
4. The LEDs operate in the programmed sequence.
5. The cycle repeats continuously without manual intervention.

Source Code

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conclusion

8. Result

The real-time traffic light control system was designed using an Arduino Uno. The red, yellow, and green LEDs operate sequentially with predefined time delays, demonstrating automatic traffic signal control using an embedded system.

9. Conclusion

Thus, a traffic light control system was designed and implemented using an embedded system. The experiment demonstrates the use of a microcontroller, digital output pins, and timing delays to automate traffic signals in a predefined sequence.

output

7. Output

The traffic light system operates in the following sequence:

```
None
RED LED ON      → 5 seconds
GREEN LED ON    → 5 seconds
YELLOW LED ON   → 2 seconds
        ↓
   Repeat Cycle
```

The LEDs switch automatically according to the programmed timing intervals.